

Chapter 02 : Power series

A series of the form,
 $a_0 + a_1(x-c) + a_2(x-c)^2 + \dots + a_n(x-c)^n + \dots$
 $= \sum_{n=0}^{\infty} a_n(x-c)^n$ is called a power series.

centered at c , where $a_n \in \mathbb{R}$ for $\forall n \in \mathbb{N}$ and $c \in \mathbb{R}$.

Example 01.

Consider the power series $\sum_{n=0}^{\infty} x^n$.

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} 1 \cdot (x-0)^n$$

Here $a_n = 1, \forall n \in \mathbb{N}$

$c = 0 \in \mathbb{R}$ The power series centered

when $x = 1$

The ps becomes $\sum_{n=0}^{\infty} 1 \cdot 1^n = \sum_{n=0}^{\infty} 1$ is divergent.

when $x = -1$

The ps becomes $\sum_{n=0}^{\infty} 1 \cdot (-1)^n = \sum_{n=0}^{\infty} (-1)^n$ is divergent

when $x = 0$

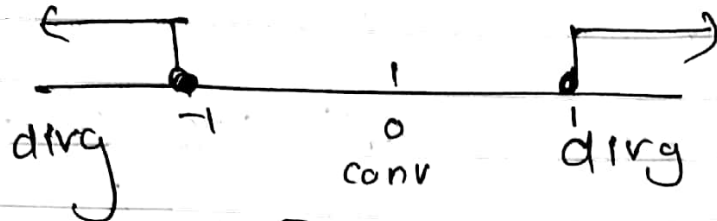
The ps becomes $\sum_{n=0}^{\infty} 1 \cdot (0)^n = \sum_{n=0}^{\infty} 0$ is convergent

when $x = 1/2$

The ps becomes $\sum_{n=0}^{\infty} 1 \cdot (1/2)^n = \sum_{n=0}^{\infty} (1/2)^n$ is convergent

A geometric series with common ratio r for $|r| < 1$. Since $|1/2| < 1$, the series is convergent.

~~The~~ In fact $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots + x^n + \dots$
 Geometric Series with common ratio x Hence
 the ps $\sum_{n=0}^{\infty} x^n$ is convergent when $|x| < 1$ and
 divergent when $|x| \geq 1$



$(-1, 1)$ The interval of convergence

$R = 1$ Radius of convergence

Example 02

Consider $\sum_{n=0}^{\infty} (n+1)^2 (x-3)^n$

$$a_n = (n+1)^2 \quad \& \quad c = 3$$

This is a power series centered at $c=3$

When $x = 3$

The ps becomes $\sum_{n=0}^{\infty} 0$ is convergent

When $x = 2$

The ps becomes $\sum_{n=0}^{\infty} (n+1)^2 (-1)^n$ is not convergent

*The convergent of a power series depends on x & a_n

* Summary

Theorem 01

If $\sum_{n=1}^{\infty} a_n$ is convergent,

the $\lim_{n \rightarrow \infty} a_n = 0$ but

if $\lim_{n \rightarrow \infty} a_n \neq 0$ we can't say $\sum_{n=1}^{\infty} a_n$ is convergent

Test for divergent

$$\lim_{n \rightarrow \infty} a_n \neq 0$$

+ $\sum_{n=1}^{\infty} a_n$ is divergent

p-series test

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

$p > 1 \Rightarrow$ the series is convergent

$p \leq 1 \Rightarrow$ The series is divergent

Comparison test

$\sum_{n=1}^{\infty} a_n$ is a given series & $\sum_{n=1}^{\infty} b_n$

there exists $\sum_{n=1}^{\infty} b_n$ $a_n \leq b_n$ &

if $\sum_{n=1}^{\infty} b_n$ is convergent

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is also convergent.

(2) If the given series is $\sum_{n=1}^{\infty} b_n$ & there

exists $\sum_{n=1}^{\infty} a_n$ s.t. $a_n \leq b_n$ &

if $\sum_{n=1}^{\infty} a_n$ is divergent

$\Rightarrow \sum_{n=1}^{\infty} b_n$ is also divergent

(*) $a_n \leq b_n$ ^{given} exist

If $\sum b_n \rightarrow$ convergent

$\Rightarrow \sum a_n \rightarrow$ convergent

~~$\sum a_n$ conver~~

$a_n \leq b_n$

$\sum a_n \rightarrow$ divergent

$\Rightarrow \sum b_n \rightarrow$ divergent

$\sum b_n \rightarrow$ divergent

X

Limit form of Comparison Test

(1) $\sum_{n=1}^{\infty} a_n$ is the given series.

if there exist $\sum_{n=1}^{\infty} b_n$ as

$a_n \leq b_n$ & $\sum_{n=1}^{\infty} b_n$ is divergent

$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L \rightarrow$ finite (+ve) value.

② $\sum_{n=1}^{\infty} b_n$ is the given series if there exists ~~a_n~~ $\sum_{n=1}^{\infty} b_n$ as $a_n \leq b_n$ & if $\sum_{n=1}^{\infty} a_n$ is convergent

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L \quad (L > 0 \text{ finite value})$$

$$\Rightarrow \sum_{n=1}^{\infty} b_n \text{ is convergent}$$

* Show that the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 4}$ is convergent.

I using the comparison test.

~~$$\sum_{n=1}^{\infty} a_n$$~~

$$a_n = \frac{1}{n^2 + 4}$$

$$b_n = \frac{1}{n^2}$$

$$\frac{1}{n^2 + 4} \leq \frac{1}{n^2}$$

~~$$\lim_{n \rightarrow \infty} \frac{1}{n^2}$$~~

~~$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$~~

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \text{ is}$$

$$a_n = \frac{1}{n^2}$$

$$\forall n \in \mathbb{N}$$

$$\frac{1}{n^2+4} < \frac{1}{n^2}$$

$\uparrow$ $\uparrow$
 a_n b_n

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

Using the p-series test

$$p=2 \quad p>1$$

The series is convergent

$\sum_{n=1}^{\infty} b_n$ is convergent

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent by comparison test.

II Using the limit form of the comparison test

$$\lim_{n \rightarrow \infty}$$

$$\frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{1+4/n^2}$$

$$\lim_{n \rightarrow \infty}$$

$$\frac{1/n^2+4}{1/n^2}$$

this is > 0 ve.
finite value

$$\lim_{n \rightarrow \infty}$$

$$\frac{n^2}{n^2+4}$$

$\Rightarrow$ ~~the~~ $\in (0, \infty)$

Since $\sum_{n=1}^{\infty} b_n$ is convergent, by the limit form of comparison test

$$\Rightarrow \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{n^2+4} \text{ is convergent}$$

~~Let~~

$$\frac{1}{5} + \frac{\sqrt{2}}{7} + \frac{\sqrt{3}}{9} + \dots \dots \dots \text{for } n \geq 2$$

Let $a_n = \frac{\sqrt{n+1}}{2n+1}$

~~Let~~

$$\frac{\sqrt{n+1}}{2n+1} < \frac{\sqrt{n-1}}{2n+1} < \frac{\sqrt{n}}{2n} < \frac{1}{2\sqrt{n}\sqrt{n}}$$

$$\frac{\sqrt{n-1}}{2n+1} < \frac{1}{\sqrt{n}}$$

$\downarrow$ $\downarrow$
 a_n b_n

$$\sum_{n=1}^{\infty} b_n < \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} < \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$$

By the p-series test we can say

$$p = \frac{1}{2} < 1$$

$$\Rightarrow \sum_{n=1}^{\infty} b_n \text{ is divergent}$$

Using the L.P of the C.T

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n-1}}{2n+1} \cdot \frac{1}{\sqrt{n}}$$

$$\frac{\sqrt{n-1}}{2n+1} \times \sqrt{n}$$

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n} \sqrt{n-1}}{2n+1}$$

$$\lim_{n \rightarrow \infty} \frac{(n)^{1/2} \times (n-1)^{1/2}}{2n+1}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^{1/2}} \times \frac{1}{n^{1/2}}$$

$$\lim_{n \rightarrow \infty} \frac{(n(n-1))^{1/2}}{2n+1}$$

$$\frac{(n^2-n)^{1/2}}{2n+1} \cdot \frac{(n^2-n)^{1/2}}{(n(n-1))^{1/2}}$$

$$\lim_{n \rightarrow \infty} \frac{(1 - \frac{1}{n})^{1/2}}{2 + \frac{1}{n}}$$

$$\frac{\sqrt{1-0}}{2+0} = \frac{1}{2} \in (0, \infty)$$

So by the Limit Form of the comparison test

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent

$\Rightarrow \sum_{n=1}^{\infty} \frac{\sqrt{n-1}}{2n+1}$ is divergent

Proof of the Comparison test

a) Suppose that $\sum_{n=1}^{\infty} b_n$ converges
 where $b_n \geq 0 \quad \forall n \in \mathbb{N}$ Let
 $b_n \geq a_n \geq 0 \quad \forall n \in \mathbb{N}$

Take any $\epsilon > 0$

Then ~~the~~ by the Cauchy ~~seq~~ criterion
 for series

$\exists N(\epsilon) \in \mathbb{N}$

$$\text{s.t. } |b_n - b_m| < \epsilon$$

$$|b_{n+1} + b_{n+2} + \dots + b_m| < \epsilon \quad \text{--- (1)}$$

whenever $m > n > N(\epsilon)$

Note: Cauchy criterion for

If sequence (x_n) converges then it
 satisfies the Cauchy criterion. Hence by
 the Cauchy criterion for series

for $\epsilon > 0$, there exist N s.t

$$|x_n - x_m| < \epsilon \quad \forall n, m \geq N$$

$$|b_n - b_m| < \epsilon \quad n < m$$

$$\cancel{b_1 + b_2 + b_3 + \dots + b_n} - (\cancel{b_1 + b_2 + b_3 + \dots + b_n} + b_m) = -b_m$$

$$|b_{n+1} + b_{n+2} + b_{n+3} + \dots + b_m|$$

Since $b_n \geq 0 \quad \forall n \in \mathbb{N}$

$$\cancel{b_{n+1}} + |b_{n+1} + b_{n+2} + \dots + b_m|$$

$$\cancel{b_{n+1}} = b_{n+1} + b_{n+2} + \dots + b_m$$

Also since $a_n \leq b_n$ for $\forall n \in \mathbb{N}$ we have

$$a_{n+1} + a_{n+2} + \dots + a_m$$

$$\leq b_{n+1} + b_{n+2} + \dots + b_m$$

we then have from ①

$$a_{n+1} + a_{n+2} + \dots + a_m \leq \epsilon$$

where $n, m < N(\epsilon)$

$$|a_{n+1} + a_{n+2} + \dots + a_m| = a_{n+1} + a_{n+2} + \dots + a_m$$

Hence

$$|a_{n+1} + a_{n+2} + \dots + a_m| < \epsilon$$

whenever $m, n > N(\epsilon)$

Therefore, by the Cauchy for series $\sum_{n=1}^{\infty} a_n$
convergent

Suppose that $\sum_{n=1}^{\infty} a_n$ is divergent
we shall show that $\sum_{n=1}^{\infty} b_n$
is divergent

$$\text{Let } S_n = \sum_{k=1}^n a_k \quad \forall n \in \mathbb{N}$$

Since $a_n > 0 \quad \forall n \in \mathbb{N}$ and
(divergent)

Note:

Theorem (3)

$\sum_{n=1}^{\infty} a_n$ converges $\iff$ the sequence
(S_n) of partial sums is bounded
 $|S_n| \leq M$ where $a_n > 0 \quad \forall n \in \mathbb{N}$
 $n \in \mathbb{N}$ & M is a +ve constant.

~~by theorem (3) the~~

by theorem (3) the sequence (S_n)
is unbounded.

Let the $\sum_{k=1}^{\infty} b_k \quad \forall n \in \mathbb{N}$. Since $b_n > 0, a_n > 0$
 $\forall n \in \mathbb{N}$:

$$a_1 + a_2 + a_3 + \dots + a_n \leq b_1 + b_2 + b_3 + \dots + b_n$$

$$\Rightarrow S_n \leq t_n$$

$\Rightarrow t_n$ is unbounded as S_n is unbounded
 $\therefore \sum_{n=1}^{\infty} b_n$ is divergent

Proof of limit form of the Comparison test

* Convergence Case

Let $\sum b_n$ be a convergent series

By the definition of the convergence, the sequence of partial sum

$$S_n = \sum_{k=1}^n b_k \text{ converges}$$

to some finite limit say S . a_n approaches infinity. Now, since a_n approaches L as n approaches infinity for any $\epsilon > 0$

$\exists$ an $N(\epsilon)$ s.t

$$\left| \frac{a_n}{b_n} - L \right| < \epsilon \quad \forall n > N(\epsilon)$$

$$- \epsilon < \frac{a_n}{b_n} - L < \epsilon$$

$$L - \epsilon < \frac{a_n}{b_n} < L + \epsilon \quad \text{--- (1)} \quad \text{Let's take } (L - \epsilon)b_n$$

$$(L - \epsilon)b_n < a_n < (L + \epsilon)b_n \quad \text{by (1) \&}$$

$$(L + \epsilon)b_n \quad \text{by } a_n, \text{ Since } \sum b_n$$

is convergent

$\sum c_n$ & $\sum d_n$ also convergent

$$① \Rightarrow c_n \leq b_n \leq d_n$$

Hence, $\sum a_n$ also converges by the comparison test.

Note

$\sum_{n=1}^{\infty} a_n$ is convergent then

$\Rightarrow \sum c a_n$ is convergent: where c is the constant.

Divergence case

Let b_n be a divergent series. Similar to the convergence case, we have

$$(L - \epsilon) b_n < a_n < (L + \epsilon) b_n$$

for $n > N(\epsilon)$

where $N(\epsilon)$ is some +ve integer.

If $\sum a_n$ were to converge

$\sum (L - \epsilon) b_n$ & $\sum (L + \epsilon) b_n$ would converge

$\Rightarrow \sum b_n$ converges

But this contradicts the assumption that b_n is divergent

$\sum b_n$ is divergent

$\Rightarrow \sum a_n$ must also be divergent

Integral test

Let f be

(i) non-negative, $f(x) \geq 0$

(ii) decreasing, $f'(x) < 0$

(iii) continuous, $[M, \infty)$, $M \geq 1$
 $\uparrow$
 $\in \mathbb{R}$

Then $\sum_{n=M}^{\infty} f(n)$ is convergent iff

$\lim_{n \rightarrow \infty} \int_M^n f(x) dx$ exists

or

$\lim_{n \rightarrow \infty} \int_M^n f(x) dx \in \mathbb{R}$

eg:- Using the integral test that
the infinite series

$\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ is divergent

①, ②, ③ ଚିତ୍ତିକାରୀ ଦେଖନ୍ତୁ.

$$\lim_{n \rightarrow \infty} \int_n^{2n} \frac{1}{x \ln x} dx$$

② $\lim_{h \rightarrow \infty} \int_2^h f(x) dx$

$$\lim_{n \rightarrow \infty} \int_2^n \frac{1}{x \ln x} dx$$

$$\lim_{h \rightarrow \infty} \int_2^h \frac{1/x}{\ln x} dx$$

$$\lim_{h \rightarrow \infty} \left[\ln[\ln(x)] \right]_2^h$$

$$\lim_{h \rightarrow \infty} \left[\ln[\ln(h)] - \ln[\ln(2)] \right]$$

$\alpha \notin \mathbb{R}$

By the integral test this series is divergent

① Let $f(x) = \frac{1}{x^2 \ln x}$, $x \geq 2$

clearly

(i) $f(x) > 0 \quad \forall x \in [2, \infty)$

(ii) $f(x)$ is a decreasing function

$\forall x \in [2, \infty)$ (Since harmonic function)

(iii) $x \in [2, \infty)$, $f(x)$ is continuous.

③ By Using Integral test.

$\Rightarrow \sum_{n=2}^{\infty} f(n)$ is divergent

$\Rightarrow \sum_{n=2}^{\infty} \frac{1}{n^2 \ln n}$ is divergent

Ratio test

Let $\sum_{n=1}^{\infty} a_n$ be infinite series with ~~not~~ non negative terms

a) If $\left| \frac{a_{n+1}}{a_n} \right| < r$ where $r < 1$ & r is a positive integer,

$\Rightarrow$ Then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent

Note

If a series absolutely converges then it is convergent

$$\sum_{n=1}^{\infty} |a_n| < \infty \quad \not\Rightarrow \sum_{n=1}^{\infty} a_n$$

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent

b) If $\left| \frac{a_{n+1}}{a_n} \right| \geq r$ where $r > 1$ &

(+) re integer

$\forall n \in \mathbb{N}$
 $\Rightarrow$ Then ~~series~~ $\sum_{n=1}^{\infty} a_n$ series is divergent

Limit form of the Ratio test

Let $\sum_{n=1}^{\infty} a_n$ be the series with $a_n > 0$

$$\text{Let } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L \in \mathbb{R}$$

a) If $L < 1 \Rightarrow \sum a_n$ is absolutely
cgt

$\Rightarrow \sum a_n$ is cgt

b) if $L > 1 \Rightarrow \sum a_n$ is divergent

c) if $L = 1 \rightarrow$ the test is inconclusive